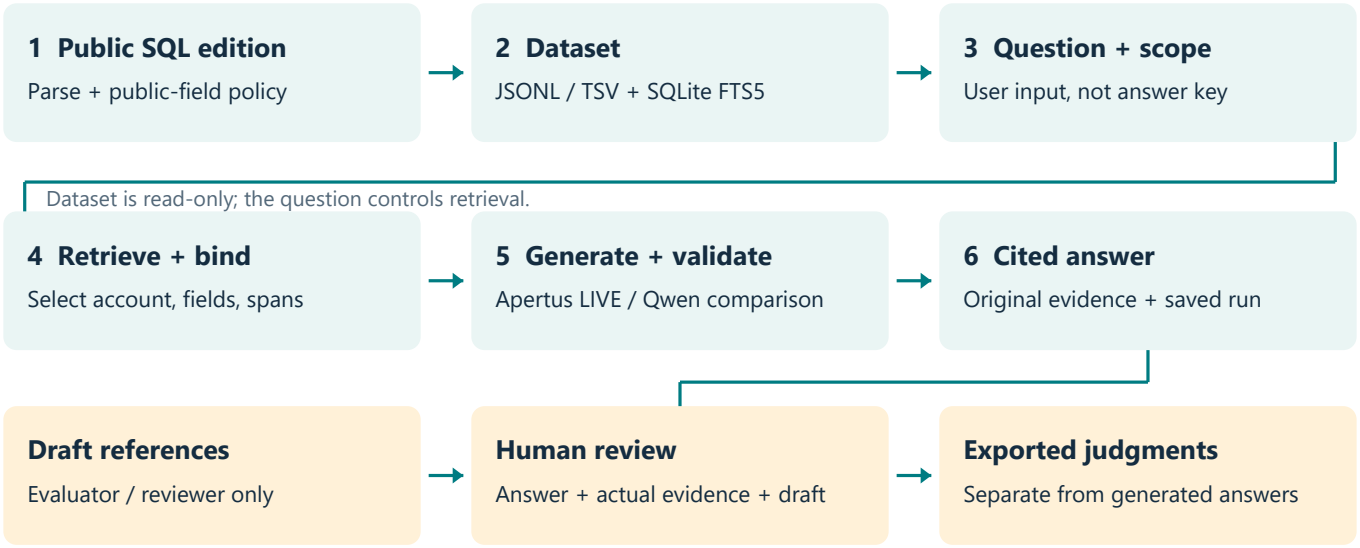
One pipeline, two separate pages

A text-only, evidence-linked application for the edited **1799 Stapfer school survey**. It makes the scholarly edition easier to search; it does not replace the edition, perform handwriting recognition, or establish that the original dataset is inaccurate.



Page	Purpose	Does it call a model?
LIVE 127.0.0.1:17860	Ask a new question; inspect its cited original evidence.	Yes, when **Ask** is clicked.
Saved run review 127.0.0.1:17863	Read recorded outputs and save separate judgments.	No. No GPU is needed.
DEMO	Labelled display fixtures for interface checks.	No; not measured model results.

Start with the application, not a benchmark rerun

```
uv run --locked python -m scripts.setup
```

After cloning the private repository and installing uv and Docker, this command verifies the included data archive, creates your private configuration and launches both web pages. You supply an Apertus endpoint and a public Hugging Face OAuth client ID. It does not download model weights, start a GPU server or generate answers.

> **Runtime / evaluation boundary:** only the question and optional transcription scope enter from a selected benchmark example. Draft answers, expected status, gold source IDs and human corrections never become runtime instructions or retrieval hints.

Native launch after setup

```
uv run --locked python app.py --port 17860
uv run --locked --group review-auth python review_app.py \
  --runs-root runs --allow-evaluation --port 17863
```

Use separate terminals. Python 3.11.16 and uv.lock pin the environment. The web pages use CPU; LIVE additionally needs a reachable model. Export reviews before restarting.

Models and reproducible inference settings

LIVE uses **apertus-qualified + bound**; the separate comparator uses **qwen-only + bound**. Each chosen model handles routing and answering. There is no deployed hybrid, and Qwen receives text only.

Setting	Apertus candidate	Qwen comparator
Model	swiss-ai/Apertus-v1.5-8B	Qwen/Qwen3-VL-32B-Instruct
Weight precision	BF16; no CPU weight offload	BF16; no CPU weight offload
Request temperature	0	0
Context / concurrency	8,192 tokens / 1 sequence	8,192 tokens / 1 sequence
Execution	Eager; vLLM generation defaults	Eager; vLLM generation defaults
Reviewed GPU fraction	0.40	0.85; sequential comparison
Default local API	127.0.0.1:8010/v1	127.0.0.1:8012/v1
Recorded request timeout	120 seconds	120 seconds in the frozen runs
Output ceilings	Scoped router 384; answer 1,024	Same; entity router 180 for both
Per-question limits	At most 3 generation POSTs; 1 shared answer-validation retry	Same limits

The client sets **temperature=0**, stage token ceilings and a **strict JSON Schema**, but not top_p, top_k, seed or a reasoning toggle. Temperature zero does not guarantee identical outputs across servers. Default Qwen timeout: 180 seconds; frozen runs: 120 seconds.

Checkpoint and server identity

Apertus revision: `a411d838600baf0e3635a3daf66fb7c55fc97bb6`. Both comparisons used the pinned Swiss AI vLLM image below. Qwen's local 14-shard checkpoint passed metadata/availability checks, not a new full-weight hash audit.

```
ghcr.io/swiss-ai/vllm_apertus_1.5_release@sha256:
d85533f251c8272218ae2e519510836376e5a335b44f2204f193018e37753bc6
```

> **Do not copy the reviewed VRAM reservations blindly.** The portable optional Apertus setup defaults to 0.90 on an available GPU, not the historical 0.40. The 0.40 + 0.85 profiles cannot coexist on one GPU. Check capacity; never stop another job to make room.

Recorded, assisted human-review results

Candidate	Development	Evaluation	Total
Apertus 8B	17 / 20	21 / 30	38 / 50 (76%)
Qwen 32B	18 / 20	29 / 30	47 / 50 (94%)

Same 50 questions, 100 judgments. Draft references, exposed splits and two reviewers limit generalization. Two separately attributed Jean/assistant amendments change Apertus's preserved original 40/50 to 38/50. No inference was rerun for this guide.

From the SQL edition to the dataset

Input: `stapferappdb_final.sql.zip`, containing `stapferappdb_final_10.5.16.sql`. The importer verifies the pinned source bytes and parses an allowlisted SQL-dump grammar. **It never executes the source SQL.**

Step	Transformation and output
1. Verify + parse	Read 27 source tables. Preserve original IDs and narrative bytes; reject unsupported SQL statements instead of producing a partial import.
2. Public projection	Admit transcription release label **Internet**; allowlisted fields only. Exclude modern contributors, internal comments, administration and binary timestamps.
3. Reusable exports	Seven JSONL record types + readable answers.tsv; separate normalized search text. Keep spelling, null/empty values, editorial markers and historical units.
4. Read-only search	SQLite FTS5 plus relational IDs. NFC/whitespace normalization for search; unicode61 tokenization with diacritic removal. Displayed original text is not modernized.
5. Verify + seal	Schemas, IDs, relationships, serialized text roundtrips, TSV/index agreement and checksums. _SUCCESS.json marks completion; failure leaves _FAILED.json.

Measured corpus, not one question-answer training file

Record unit	Exported count	Important qualification
Transcription / question	2,250 / 60	Transcriptions are not answer rows.
Answer rows	92,524	176 null + 12 blank retained; 92,336 indexed.
Schools / places	3,229 / 2,143	School links describe transcription context.
Teachers / school-teacher links	2,443 / 3,217	Not proof that an answer belongs to one school.

14,646 answer rows retain an uninterpreted school-assignment integer. No ownership is invented. Missing verified PDF-page mappings remain null; an age or editorial page marker cannot be used as a PDF page index.

```
uv run --locked python -m core.dataset_cli build \
  --source data/raw/stapferappdb_final.sql.zip \
  --out data/derived/j2-full
uv run --locked python -m core.dataset_cli verify \
  --data-dir data/derived/j2-full
```

The included bootstrap tar already supplies this completed projection. Rebuild only into a **new** output directory; existing output is never overwritten. JSONL/TSV, SQLite, schema, policy, source manifest, transformations and validation report remain together.

Distribution, provenance and rights

The verified 44.8 MB gzip tar supplies the dataset, four frozen runs and original review exports (2,119 files). It excludes raw SQL, weights, credentials and new account reviews. Setup rejects unsafe paths and mismatched existing files; historical run snapshots stay unchanged. Source manifests, transformation logs, run seals and raw request/response events preserve the audit trail.

MIT covers our code, not model weights or the scholarly edition. Preserve original IDs, spelling, markup, units and archival references. See the Stapfer-Enquête catalogue and open-use conditions. A teacher's testimony and an AI interpretation remain distinct evidence layers.

Retrieval, account selection and field binding

RAG means retrieving evidence before generating an answer. Here it is **SQLite lexical search plus relational account/field selection**, not vector embeddings, a trained retriever or an LLM reading the whole archive in one request.

Question scope	How records are selected
Transcription + one explicit code	Example: transcription 16 and I.1. Software retrieves that source field directly; a routing model call is unnecessary.
Transcription, natural question	Split literal requested details into r1, r2, etc. The model selects one catalogue field per detail, using the full question and original/French field definitions.
Several explicit codes	Retrieve the caller-supplied fields first. Their actual source passages can assist routing; a field title may omit details contained in the historical response.
No transcription ID	Model copies place/person names from the question and selects up to 3 fields. Match names in primary identity answers; require a unique common transcription, then fetch its related fields.

Why selecting the right document is not enough

A teacher's own children and a school's pupils are different populations, even if both fields contain a real number. Each r-key is bound to a specific field and permitted citation spans. A model must answer every key using its own evidence; an empty binding does not expand into all available passages. The literal splitter and model's field choice are fallible, not a semantic completeness certificate.

Bounded evidence, with visible uncertainty

Identical source text is grouped only within one transcription while retaining every included ID. Limits: **5 passage groups, 2,000 characters per passage and an 11,000-character serialized packet**. Omitted IDs and whether a passage is complete are recorded. A separate conservative 18,000-character message guard is not an exact token measurement; the server enforces its 8,192-token context.

For entity-based selection, literal name matching can supply an anchor if the model supplies none. This is source-grounded lexical matching, not verified historical alias resolution. Ambiguous accounts, missing required fields or incomplete entity packets cause a hold rather than an invented answer.

> **The useful 'hints' are real source context:** the catalogue definitions, user-supplied scope, original passages and typed requested-detail checklist. They are not the draft reference answer, expected status, reviewer correction or a gold record-ID lookup.

Where the draft references come from: a separate, assistant-prepared benchmark snapshot pairs questions with proposed facts/statuses and quotations checked against the edition. The RAG does not create that answer key during a query. Source matching verifies quotation provenance, not every translation, interpretation or question's clarity; human review must still assess those points.

Archive-wide totals, broad comparative history, unusual spellings and multilingual paraphrases are not guaranteed by this bounded account-selection method. 'No useful evidence retrieved' means a limit of this search, not proof that the archive has no answer.

What the model actually receives

Every completion sends: **model + temperature 0 + max_tokens + messages + strict response_format JSON Schema** to /v1/chat/completions. Messages contain a system policy and a user JSON payload. The policy requires source-only answers, all requested details, preserved qualifiers/terms, supported arithmetic and abstention when evidence is unresolved.

Actual saved DEV-01 user-message JSON, compacted without changing any field/value. Both frozen models received this payload. I.1 plus transcription 16 bypassed routing; no expected answer was supplied.


```
{
  "question": "According to I.1, what is the place called?",
  "transcription_id": 16,
  "requested_details": {"r1": {
    "detail": "what is the place called?",
    "question_codes": ["I.1"],
    "citation_spans": [{
      "span_id": "s1", "record_id": "stapfer:answer:41",
      "question_code": "I.1",
      "quote": "Hörndli, heist der Ort, wo die $ Schull $$ ist.",
      "survey_question": "Name des Ortes, wo die Schule ist.",
      "survey_question_french": "Nom de l’endroit où est située l’école.",
      "passage_complete": true, "pdf_page_index": null
    }]
  }},
  "complete_schedule_checks": {
    "active": false, "currency_amounts": [], "period_terms": [],
    "not_population_pairing_or_entailment_verification": true
  },
  "school_links_are_context_not_ownership": true,
  "evidence_is_not_a_complete_census": true
}
```


The response schema permits only answered or insufficient_evidence. For answered, each r-key needs text and 1-3 citations selected from its own span IDs; message is empty. The actual model output here was:


```
{"status":"answered","answers":{"r1":{"text":"Hörndli",
 "citations":[{"span_id":"s1"}]}},"message":""}
```


Software maps s1 back to the original quote and record ID, creates one stored claim and displays **Hörndli [stapfer:answer:41]**. Source metadata is copied from the dataset, not invented by the model: BAR B0 1000/1483, Nr. 1421, fol. 228-229v; PDF page unverified.

> Validation rejects missing r-keys, foreign-field citations, malformed outputs and non-normal finishes. Complete-fee checks conserve source amounts/units/period terms, not their meaning. A permitted span ID or exact quotation is not proof that the answer follows from it.

Private configuration and implementation

Keep .env (model/data), .env.review-oauth (public client ID and private session secret), .env.compose (mounts/network) and optional .env.apertus (GPU server) out of Git. The registered HF callback must exactly match this review installation, normally http://127.0.0.1:17863/auth/huggingface/callback. Review OAuth does not authorize gated model downloads. README gives the separate checkpoint/server setup; NVIDIA/BF16 serving requirements do not apply to saved-run review.

Implementation: core/answer_binding.py (routing, prompt, schema); core/apertus.py (wire request); core/entity_retrieval.py and evidence.py (selection/binding); evaluation*.py and review_app.py/ui (records/review). Checked baseline `db7cbde576ed`, plus frozen bound-03 requests.

Human review: states, claims and persistence

State family	State	Meaning
Backend	answered	A response passed the mechanical contract. UI: answer produced, not human-reviewed. It can still be wrong.
Backend	insufficient_evidence	No answer is asserted under this evidence/scope/budget. Can be a valid abstention or a false abstention.
Backend	error	Request/configuration/transport/validation failure. Not evidence that a historical fact is absent.
Human decision	Pending / null	Not reviewed; no accuracy claim.
Human decision	Correct	All requested details are correct and every stored claim is supported; an appropriate abstention can also be correct.
Human decision	Incorrect	Wrong meaning, missing requested detail, unsupported claim or inappropriate abstention.
Human decision	Uncertain	Reviewer cannot decide. Record why; do not count it as confirmed correct or incorrect.

What exactly is a supported claim?

The count is the number of supported entries in the saved `claims[]`, not the number of citations, sources, nouns or sentences. One claim entry can contain several facts; count it supported only if the delivered evidence supports **all** of them. Do not split a grouped claim retrospectively to raise the score.

If three books appear inside **one** saved claim, check all three books, then count 1/1 or 0/1. If there are **three** saved claim entries, the range is 0-3. A true duration-only claim can be supported 1/1 yet the whole answer Incorrect if requested weekdays were omitted. An abstention has 0 claims; zero supported claims is not the same as a failed abstention. A blank/Not counted support field means unassessed, not zero.

Review workflow

1. Select the exact saved run/case. Read the question and generated answer.
2. Check every stated fact against the **evidence actually delivered**, then compare the draft required facts/quotations. A draft can be incomplete or ambiguous; flag it rather than treating it as approved truth.
3. Set decision + supported count; add notes explaining any gap. A human suggestion is separate from the original answer, never a replacement.
4. **Save** keeps edits in page memory. **Export** writes persistent files. Export before closing, reloading, changing runs or restarting; use Resume to restore.

Hugging Face identifies the signed-in account, not historical expertise or institutional affiliation. New accounts review in separate writable exports; Jean/Jeni originals remain read-only. Exports bind to the saved run/response. Seeing drafts or original decisions means this is assisted review, not a certified blind protocol.

Do not claim the organizer's document target from these scores. The separate 100-document audit is still 0/100 reviewed in these records. Its checks target at least 80% correctly processed documents and less than 5% structural errors on that sample; question accuracy does not establish either rate, corpus representativeness or an at-most-8-GB deployment. Report retrieval, citation integrity, human correctness and latency separately.